Miguel Masó, PhD

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City: Barcelona (08006) Birth date: 9/4/1989 ORCID: 0000-0003-0962-8550

Employment history

2022 - today	**SOCOTEC Spain** - Buildings and cities Head of development and innovation
2022 - today	**Universitat Politècnica de Catalunya (UPC)** - Barcelona School of Civil Engineering Adjunct lecturer
2017 - 2022	**International Centre for Numerical Methods in Engineering (CIMNE)** Doctoral student with FPI grant

Education

2017 - 2022	**PhD in Civil Engineering** UPC, Barcelona, Spain
2013 - 2015	**Bachelor's degree in Philosophy and Theology** Ponitificia Universtià della Santa Croce, Rome, Italy
2007 - 2013	**Ingeniero de Caminos, Canales y Puertos** (1st and 2nd cycle studies) UPC, Barcelona, Spain

Languages

Native Spanish - Native Catalan - C1 English - B1 Italian

CV Summary

After studying civil engineering and a period studying philosophy and theology in Rome, he began a doctorate in civil engineering at CIMNE. During the initial years of his doctoral studies, he combined research with work for social purposes at the Montseny Foundation.

Durign the research period at CIMNE, he worked on the stabilization of finite elements for the hyperbolic differential equations, with focus on numerical implementation, multi-physics coupling, and multi-scale resolution. These developments have been succesfully applied to various projects. In addition to the finite element stabilization, he also researched on the particle finite element method, either with moving mesh and fixed mesh.

Currently, he is working at the Building and Cities department of Socotec-Spain. He has participated in several international projects for structural design, taking on responsibilities of vibration assessment and plasticity. He is leading the innovation and development department.

Publications

Journal papers

[1] Miguel Masó, Ignasi De-Pouplana, and Eugenio Oñate. "A FIC-FEM procedure for the shallow water equations over partially wet domains". In: *Computer Methods in Applied Mechanics and Engineering* 389 (Feb. 2022), p. 114362. DOI: 10.1016/j.cma.2021.114362.

[2] Miguel Masó et al. "A Lagrangian-Eulerian procedure for the coupled solution of the Navier-Stokes and shallow water equations for landslide-generated waves". In: *Advanced Modeling and Simulation in Engineering Sciences* (July 2022), pp. 2213–7467. DOI: 10.1186/s40323-022-00225-9.

[3] A. Montanino, M. Masó, and A. Franci. "One-way coupled Eulerian-Lagrangian strategy for waves propagation and impact on coastal areas". In: *Ocean Engineering* (2024). [*Under review*].

[4] Ignasi De-Pouplana et al. "A prototype of a micro-scale model for the distribution of NO2 in urban areas". In: *Atmospheric Pollution Research* 14.2 (2023), p. 101668. ISSN: 1309-1042. DOI: 10.1016/j.apr.2023.101668.

[5] A. Puigferrat et al. "Semi-Lagrangian formulation for the advection-diffusion-absorption equation". In: *Computer Methods in Applied Mechanics and Engineering* 380 (July 2021), 113807:1–113807:26. DOI: 10.1016/j.cma.2021.113807.

Theses

[6] M. Masó. "Coupling shallow water models with three-dimensional models for the study of fluid-structure interaction problems using the particle finite element method". Universitat Politècnica de Catalunya, 2022.

Conferences

[7] M. Maso, I. De-Pouplana, and E. Oñate. "Coupling shallow water models with 3D models for the study of fluid-structure interaction problems using the particle finite element method". In: *Particles 2019 - VI International Conference on Particle-based Methods*. Barcelona, Spain, 2019.

[8] M. Maso et al. "A one-way coupled Lagrangian-Eulerian procedure for the solution of landslide-generated waves". In: *WCCM APCOM - 15th World Congress on Computation Mecchanics & 8th Asian Pacific Congress on Computation Mechanics*. Yokohama, Japan (Virtual), 2022.

[9] M. Maso et al. "Un Procedimiento Lagrangiano-Euleriano para la Solución Acoplada de las Ecuaciones de Navier-Stokes y Aguas Poco Profundas para Olas Generadas por Deslizamientos de Laderas". In: *CMN XV - Congress on Numerical Methods Engineering*. Las Palmas de Gran Canaria, Spain, 2022.

Stays and other activities

Guest. Centro de Investigación en Matemáticas. Purpose: Research. Comparable tasks: WP3 of project TCAiNMaND-PIRSES-GA-2013-612607 Numerical Methods for Real Time Computations Research on stabilized formulations the Finite Element Method applied to hyperbolic conservation laws. 18/07/2017 - 19/10/2017.

Teaching activities

2022 - today	**Structural Dynamics.** Credits: 7'5. Degree: Master's degree in Structural & Construction Engineering. UPC.
2022 - 2022	**Structural Analysis.** Credits: 7'5. Degree: Master's degree in Civil Engineering. UPC.
2017 - 2017	**Numerical Methods.** Credits: 6. Degree: Bachelor's degree in Materials Engineering. UPC.

Participation in R+D+i projects

Title: (2021 SGR 01349) Mecànica de Medis Continus i Estructures
Financing entity: AGAUR. Agència de Gestió d'Ajuts Universitaris i de Recerca
From: 01/01/2022 To: 30/06/2025
PI: Cervera Ruiz, Luis Miguel

Title: (DPI2015-67857-R) Proyectos de I+D: Retos de la Sociedad 2015
Financing entity: MINECO
From: 01/01/2016 To: 31/12/2018
PI: Codina, Ramon; Baiges, Joan

Title: (FP7- 612607) FP7-PEOPLE-2013-IRSES
Financing entity: EC
From: 01/01/2014 To: 31/12/2017
PI: Larese de Tetto, Antonia

Miscellaneous

- **Skills:** Computational Mechanics, Fire Engineering, Ocean Engineering, Vibration Assessment
- **Programming:** C++, Python, C#, .NET
- **Tools:** Robot Structural Analysis API, Revit API
- **Miscellaneous:** ENVISION SP 2023-2024